WEEK 8 – PROJECT PROGRESS REPORT

University Bus Tracking System

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Course / Programme	**MCA – Cyber Security**
Week	**Week 8**

1. Project Objective

- To extend the Admin Module by implementing Bus Management backend APIs, bus photo upload functionality, and integrating live backend data with the Admin Dashboard.

2. Work Completed

Admin Authentication Integration

- Integrated JWT-based authentication with the Admin Dashboard.
- Verified admin login and protected API access using Bearer Token authentication.
- Resolved the 401 Unauthorized issue and verified successful API responses.

Bus Management Backend APIs

- Implemented the API to create a new bus (POST).
- Verified the API to retrieve bus records (GET).
- Implemented and tested the API to update bus details (PUT).
- Implemented a safe delete endpoint with linked-trip checks. The delete operation itself was not executed on existing data.

Bus Photo Upload

- Implemented front-side and side-view bus photo upload.
- Added file validation for JPG, PNG and WEBP formats.
- Saved uploaded images with unique filenames in the backend uploads directory.
- Configured FastAPI static file serving and verified the uploaded image URLs.

Admin Dashboard Integration

- Connected Overview, All Users, Buses, Drivers and Routes sections to existing backend APIs.
- Verified successful data loading from the protected APIs.

Routes & Stops

- Integrated existing route data into the dashboard in read-only mode.

3. Testing and Verification

- Admin login and JWT authentication were tested successfully.
- Bus Update API was tested successfully.
- Bus Photo Upload API and image URLs were tested successfully.
- Dashboard API requests for Overview, Buses, Drivers and Routes returned HTTP 200 OK after authentication was refreshed.
- Existing bus and related database records were preserved.

4. Pending Tasks

- Connect the Add Bus, Edit Bus and Delete Bus controls in the Admin Dashboard UI.
- Display uploaded bus photos in the dashboard.
- Implement Add Route and Add Stop functionality.
- Continue Live GPS Tracking integration.
- Complete the remaining Admin Module integrations.

5. Outcome

- The Admin Dashboard is connected to the existing backend APIs. Bus Management backend endpoints and bus photo upload have been implemented and tested. The project is progressing towards complete Admin Module integration.

6. Next Week's Plan

- Connect Add Bus, Edit Bus and Delete Bus UI with the backend APIs.
- Show front and side bus photos in the Admin Dashboard.
- Continue the remaining Admin Module and Live GPS Tracking work.